

CENG381 Stochastic Processes

Gambler's Ruin - Question Set 1

Notation used in this question set:

N	total capital in the game (upper boundary)
k	gambler's current fortune (starting state)
R(k)	probability that the gambler reaches N before 0, starting with k
f(k)	expected number of steps until absorption, starting with k
p	probability of winning a single bet (gaining 1)
q = 1-p	probability of losing a single bet (losing 1)

Q1 (30 points). A gambler starts with $k = 3$ dollars and plays a fair coin-flip game ($p = q = 1/2$) against an opponent who starts with $N - k = 7$ dollars, so $N = 10$. At each step the gambler wins 1 dollar (prob $1/2$) or loses 1 dollar (prob $1/2$). The game ends when one player is broke.

- Write the recurrence relation for $R(k)$, the probability of the gambler reaching $N = 10$ before 0, starting with k dollars. State the boundary conditions $R(0)$ and $R(N)$.
- Solve the recurrence to find $R(k)$ for all $k = 0, 1, \dots, 10$. Use the formula for the unbiased case: $R(k) = k / N$.
- What is $R(3)$? Interpret this result in plain language: what are the gambler's chances of winning the game?

Q2 (35 points). Now consider the biased version of Gambler's Ruin with $N = 6$, starting fortune $k = 2$, winning probability $p = 1/3$, and losing probability $q = 2/3$.

- Write the recurrence $R(k) = p * R(k+1) + q * R(k-1)$ with boundary conditions $R(0) = 0$ (ruin) and $R(N) = 1$ (win). Show that the characteristic equation is $p*r^2 - r + q = 0$ with roots $r_1 = 1$ and $r_2 = q/p$.
- Since $r_1 \neq r_2$ (because $p \neq q$), write the general solution $R(k) = A * 1^k + B * (q/p)^k = A + B*(q/p)^k$. Apply the boundary conditions to find A and B.
- Compute $R(2)$ (the gambler's probability of winning from $k=2$). Is this greater or less than the unbiased value of $k/N = 2/6 = 1/3$? Explain why.

Q3 (35 points). Return to the unbiased game ($p = q = 1/2$) with $N = 8$ and starting fortune k .

- Write the recurrence for $f(k)$, the expected number of steps until the game ends (absorption at 0 or N), starting at k . State the boundary conditions $f(0) = f(N) = 0$.

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- b) Solve the recurrence. Note that the homogeneous part has repeated root $r = 1$, so the particular solution has the form $c \cdot k$. Show that the full solution is $f(k) = k \cdot (N - k)$.
- c) From which starting fortune k (among $k = 1, \dots, 7$) does the game last longest on average? Compute $f(k)$ for $k = 2, 4, 6$ and verify the formula $f(k) = k \cdot (8 - k)$.